Gemmechu Hassena

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Education

Cornell University — *Ithaca, NY*

Ph.D. in Computer Science (GPA: 4.13/4.0), Advised by Professor Bharath Hariharan and Steve Marschner. Working at the intersection of computer vision, graphics, and machine learning. My research focuses on understanding scene geometry and lighting from casually captured images. — Aug. 2021 - May, 2027

Addis Ababa University — *Addis Ababa*

BS in Software Engineering — Sep. 2016 - Oct. 2021

Publications

[1] **Gemmechu Hassena**, Jonathan Moon, Ryan Fujii, Andrew Yuen, Noah Snavely, Steve Marschner, Bharath Hariharan. "ObjectCarver: Semi-automatic segmentation, reconstruction and separation of 3D objects" 3DV 2025

[2] Weeks, B.C., Zhou, Z., O'Brien, B.K., Darling, R., Dean, M., Dias, T., **Hassena, G.**, Zhang, M., Fouhey, D.F. "A deep neural network for high throughput measurement of functional traits on museum skeletal specimens." *Methods in Ecology and Evolution*, 2022.

[3] Zhizhuo Zhou, **Gemmechu Hassena**, Brian C. Weeks, David F. Fouhey. "Quantifying Bird Skeletons" *CV4Animals Workshop, CVPR*, 2021.

Research Projects

3DGS Relighting for Outdoor Scenes Using a Video Generative Model — *LucasFilm*

Advised by Dan Goldman, Daniele Bernabei and Stephen Hill. — Ongoing

Neural relighting without shading assumptions — *Cornell*

Advised by Bharath Hariharan and Steve Marschner, we investigate how to relax shading assumptions in inverse rendering to enable relighting of diverse materials. — Ongoing

DynamicFusion with Gaussian Splatting — *Meta*

Advised by Chris Sweeney and Adam Harley. We showed how 2D tracking can be lifted to 3D and integrated into 3D Gaussian Splatting for dynamic reconstruction. — May 2025 - Aug 2025

Personalized color correction — *Cornell University*

Advised by Prof. Abe Davis, We ask how can we color correct images without white balance color cards — Aug 2021 - Dec 2021

Pointcloud denoising with laplacian smoothing — *INRIA - Ecole Polytechnique*

Advised by Prof. Maks Ovsjanikov, we worked on removing noise from noisy pointcloud with unsupervised learning. — Oct 2021 - Mar 2022

Scene understanding using humans as a ruler — *University of Michigan*

Advised by Prof. David Fouhey. We asked can we estimate the depth and the ground floor from just YouTube videos by using people's height as a reference. — May 2020 - Sep 2020

Work Experience

Lucasfilm / Disney | *Research Intern* — *San Francisco, CA*

Researched relightable 3D Gaussian Splatting for outdoor scenes, combining physically based rendering, illumination estimation, and diffusion-based refinement to enable realistic relighting from multiview images. — Summer 2026

Meta | *Research Scientist Intern* — *Redmond, US*

Worked on dynamic 3D reconstruction from multi camera video streams (Distributed Camera). — May 2025 - Aug 2025

Cornell | *Graduate Teaching Assistant* — *Ithaca*

CS6670 (Graduate computer vision) and CS 5/4780(Intro to Machine learning), for a class of 400 student with four other graduate TA I wrote the homework, Mid-term and Final exam questions. — Fall 2022 and Spring 2023

A2SV | *Team Lead in Project Management* — *Addis Ababa*

Led team of four people on the development of two software application projects — Mar 2022 - Aug 2022

INRIA: Ecole Polytechnique | *Research Intern* — *Paris*

Advised by Prof. Maks Ovsjanikov — Oct 2021 - Mar 2022

University of Michigan | *Research Intern* *Ann Arbor - Remote*

Advised by Prof. David Fouhey May 2020 - Jul 2021

Awards

Telluride scholarship: 5 year full room and boarding

Telluride Association 2025

Exemplary Service Award: Early Career Graduate Students

Cornell University 2024

Distinguished Leadership in Service award

Cornell Bowers CIS 2024

Talks

- *Invited talk,* Industrial Light & Magic (ILM) / Lucasfilm, December 2025
- *Lighting speaker,* IVY Collective: Harvard University, Boston, 2025
- *Panelist,* Summer Success Symposium, 2024
- *Lighting speaker,* Grad Cohort for Ideals, Minneapolis, 2024
- Cornell Graphics and Vision Seminar, Host Abe Davis, 2024
- Ecole Polytechnique, Host Maks Ovsjanikov, 2024
- NERD Podcast, Host Yeabsra Ashebir, 2024
- *Devfest Addis*, Addis Ababa, Host Google Developer Group, 2024
- *Lighting speaker,* NYC vision day, 2023
- *SONIC,* Cornell, Host Amy Huang, 2023
- *Panelist,* BURE, Host Eva Tadros (department chair), Cornell, 2023
- *Panelist,* African Undergraduate Research Adventure, Host Todd Austin and Valeria Bertacco, University of Michigan, 2023
- *Panelist,* McNair Scholar, Host Amy Huang, Cornell, 2023
- *Speaker,* Cornell Graphics and Vision Seminar, Host Bharath Hariharan, 2023

Skills

Software	Blender (animation, modeling, material, scripting), Photoshop
Tool/Framework	Gaussian Splatting, NeRF, COLMAP
Language	English, Oromo(Native), Amharic(Native), Arabic(Basic)

Volunteer Services

Computer Vision and Graphics Research Initiative | Founder & Lead Organizer *Addis Ababa*

- Founded and led a research program connecting students with mentors from Cornell, Stanford, Notre Dame, Google, Purdue, and more.
- Participants went on to graduate study and research opportunities at the University of Cambridge, CMU Africa, MIT summer research programs, and the Fatima Research Fellowship, published work at a CHI workshop and more.

Dec 2024 - present

Summer Geometry Initiative | Social hour chair *MIT*

Summer 2023 and 2024

Undergraduate Student Mentorship | Mentor *Cornell University*

- Mentored three undergraduate researchers, resulting in a co-authored paper. One mentee Awarded the Cornell CIS Undergraduate Research Excellence Award.

Sept 2023 - Present

PhD visit day Czar | Organizer *Cornell University*

Mar 2023

Inspire Student Club | President *Addis Ababa University*

- Organized Multiple Hackathons and Seminars
- Hosted AI in Ethiopia Conference and Jack Dorsey's visit to Addis Ababa University

Nov 2019 - Nov 202